

Pseudo-code for Transient Brokerage Simulation

1 State, Notation, and Scores

Let $[N] = \{1, \dots, N\}$ and let the broker node be $b = N + 1$. The mutable simulation state at the end of period t is

$$S^t = \left(X^t, G^t, \{\mathcal{H}_i^t, \Theta_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, \tau_i^t, M_i^t\}_{i=1}^N, \mathcal{H}_b^t, \Theta_b^t, \text{Roster}^t, D^t, \text{rep}_b^t, \kappa_b^t, n_{\kappa,b}^t \right).$$

Here M_i^t is the set of current-period counterparties, χ_i^t indicates whether agent i has ever used the broker, and τ_i^t is agent tenure. Current-period demands are discarded at the end of each period.

Table 1: Main symbols and dimensions.

Symbol	Dimension/type	Definition
<i>Core and conventions</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
T	scalar	Number of simulation periods.
b	node	Broker node, $b = N + 1$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t ; S is current state when t is clear.
omitted t	convention	Unadorned state variables inside a period refer to current-period values.
Unif, $\mathcal{N}$, Binomial	distributions	Uniform, normal, and binomial distributions.
ReLU, argsort, $\mathbf{1}\{\cdot\}$	functions	ReLU activation, sorting operator, and indicator function.
$I_d, \ \cdot\ _2, \ \cdot\ _F, \text{tr}$	operators	Identity matrix, Euclidean norm, Frobenius norm, and trace.
<i>Parameters</i>		
d	scalar	Type dimension.
d_γ	scalar	Number of active curve dimensions.
σ_x	scalar	Type-noise scale.
K	scalar	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
k_G	scalar	Watts-Strogatz graph degree.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime gain.
σ_ε	scalar	Output-noise standard deviation.
λ_c	scalar	Cost rate.
ω	scalar	Satisfaction/confidence EWMA weight.

Symbol	Dimension/type	Definition
h_a, h_b	scalars	Agent and broker hidden widths.
E_{init}	scalar	Initial training steps.
η_{lr}	scalar	MLP learning rate.
$n_{\text{strangers}}$	scalar	Period stranger-pool size.
η_{exit}	scalar	Agent exit probability.
p_{roster}	scalar	Roster-churn probability.
Δ_{net}	scalar	Network-measure interval.
$n_{\text{min}}, \kappa_{\text{max}}$	scalars	Principal-mode confidence gate.
<i>Types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{b\}$.
x_i^t	$\mathbb{R}^d$	Agent i type.
X^t	$\mathbb{R}^{d \times N}$	Type matrix $[x_1^t, \dots, x_N^t]$.
c	$\mathbb{R}^d$	Ideal type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric regime operator.
$f(x_i, x_j)$	scalar	Deterministic match signal.
$g(x_i, x_j)$	scalar	Regime gain.
q_{ij}^t	scalar	Realized output, $Q + f(x_i, x_j) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Standard broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
$\mathcal{H}_i^t$	$\mathbb{R}^{d \times n_i^t} \times \mathbb{R}^{n_i^t}$	Agent i history of partner types and realized outputs.
$\mathcal{H}_b^t$	$\mathbb{R}^{d \times n_b^t} \times \mathbb{R}^{d \times n_b^t} \times \mathbb{R}^{n_b^t}$	Broker history of ordered pair types and outputs.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$\hat{q}_i^t$	function	Agent predictor $\mathbb{R}^d \rightarrow \mathbb{R}$.
$\hat{q}_b^t$	function	Broker predictor $\mathbb{R}^{2d} \rightarrow \mathbb{R}$.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U^t	subset of $[N]$	Period stranger pool.
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
τ_i^t	integer	Agent tenure.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.

Symbol	Dimension/type	Definition
$\mathcal{R}^t$	list	Accepted records $(i, j, z, \pi, q, \hat{q}, p, o_1, o_2)$; $\pi = 1$ for principal positions and p is principal price or NaN.
<i>Metrics and sentinels</i>		
m^t	$\mathbb{R}^m$	Period metrics.
$\mathcal{M}$	$\mathbb{R}^{T \times m}$	Metric table, $\mathcal{M}_t = m^t$.
κ_b^t	scalar	EWMA broker exposure MAE.
$n_{\kappa, b}^t$	integer	Cumulative broker exposure-error count.
$\perp, \emptyset, \text{NaN}$	sentinels	Undefined score, empty object, and missing numeric value.

The matching function is

$$f(x_i, x_j) = \rho \frac{x_i^\top c + x_j^\top c}{2} + (1 - \rho) g(x_i, x_j) x_i^\top A x_j, \quad g(x_i, x_j) = 1 + \delta \text{sign}(x_i^\top B x_j).$$

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\}, \quad r = 0.60 \bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c(\bar{q}_{\text{cal}} - r).$$

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_\Theta(\xi) = w_2^\top \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$. Agent networks use $(p_{\text{in}}, h) = (d, h_a)$; the broker network uses $(p_{\text{in}}, h) = (2d, h_b)$. The symmetric broker score is

$$v_b^t(i, j) = \frac{1}{2} [\hat{q}_b^t([x_i^t; x_j^t]) + \hat{q}_b^t([x_j^t; x_i^t])].$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined only for observed graph neighbors and period strangers:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in U^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means dominate predictor extrapolations when pair-specific experience exists. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

2 Top-Level Simulation

Algorithm 1 Transient Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and metric table $\mathcal{M} \in \mathbb{R}^{T \times m}$.

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1:  $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$ 
2: for  $t = 1, \dots, T$  do
3:    $(S^t, \mathcal{R}^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$ 
4:    $m^t \leftarrow \text{COLLECTMETRICS}(S^t, \mathcal{R}^t, \vartheta)$ 
5:    $\mathcal{M}_t \leftarrow m^t$ 
6: return  $(S^T, \mathcal{M})$ 

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3 Training and Channel Choice

Local variables for training

Symbol	Dimension/type	Definition
Ξ_{tr}	$\mathbb{R}^{p_{\text{in}} \times n}$	Generic training inputs.
y	$\mathbb{R}^n$	Generic training targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Gradient-step index.
W	integer	Training-window length.
n_i, n_b	integers	Agent and broker history lengths.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker gradient-step counts.
$\Xi_{i,\text{tr}}, y_i$	matrix, vector	Agent i training batch.
$\Xi_{b,\text{tr}}, y_b$	matrix, vector	Symmetry-augmented broker training batch.

Algorithm 2 TrainNN

Require: Network parameters Θ , data $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y \in \mathbb{R}^n$, steps E , learning rate η_{lr} .

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1: for  $e = 1, \dots, E$  do
2:    $\mathcal{L}(\Theta) \leftarrow n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr},\ell}) - y_{\ell})^2$ 
3:    $\Theta \leftarrow \Theta - \eta_{\text{lr}} \nabla_{\Theta} \mathcal{L}(\Theta)$ 
4: return  $\Theta$ 

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Algorithm 3 TrainPredictors

Require: State S , period t , parameters $(E_{\text{init}}, \eta_{\text{lr}})$.

- 1: $W \leftarrow 500$
 - 2: **for** $i = 1, \dots, N$ **in parallel do**
 - 3: $n_i \leftarrow |\mathcal{H}_i|$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
 - 4: **if** $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
 - 5: $(\Xi_{i,\text{tr}}, y_i) \leftarrow$ most recent $\min(W, n_i)$ columns of $\mathcal{H}_i$.
 - 6: $E_i \leftarrow \max\{50, \lceil E_{\text{init}} \Delta_i / n_i \rceil\}$.
 - 7: $\Theta_i \leftarrow \text{TRAINNN}(\Theta_i, \Xi_{i,\text{tr}}, y_i, E_i, \eta_{\text{lr}})$; $\Delta_i \leftarrow 0$.
 - 8: $n_b \leftarrow |\mathcal{H}_b|$, $\Delta_b \leftarrow$ broker observations since last training.
 - 9: **if** $n_b > 0$ and $\Delta_b > 0$ **then**
 - 10: Build the last $n = \min(W, n_b)$ broker observations $\{(x_u, x_v, y_\ell)\}_{\ell=1}^n$.
 - 11: $\Xi_{b,\text{tr}} \leftarrow ([x_u; x_v]_{\ell=1}^n, [x_v; x_u]_{\ell=1}^n) \in \mathbb{R}^{2d \times 2n}$.
 - 12: $y_b \leftarrow (y_1, \dots, y_n, y_1, \dots, y_n) \in \mathbb{R}^{2n}$.
 - 13: $E_b \leftarrow \max\{50, \lceil E_{\text{init}} \Delta_b / n_b \rceil\}$.
 - 14: $\Theta_b \leftarrow \text{TRAINNN}(\Theta_b, \Xi_{b,\text{tr}}, y_b, E_b, \eta_{\text{lr}})$; $\Delta_b \leftarrow 0$.
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Local variables for ChooseChannel

Symbol	Dimension/type	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 4 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

- 1: $U_{\text{self}} \leftarrow s_{i,\text{self}}$.
 - 2: $U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1 - \chi_i) \text{rep}_b$.
 - 3: **if** $U_{\text{self}} > U_{\text{broker}}$ **then**
 - 4: **return** self.
 - 5: **if** $U_{\text{broker}} > U_{\text{self}}$ **then**
 - 6: **return** broker.
 - 7: **return** self w.p. $1/2$, else broker $\triangleright U_{\text{self}} = U_{\text{broker}}$.
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4 Initialization

Local variables for Initialize

Symbol	Dimension/type	Definition
ν_ℓ	integer	Frequency for active curve coordinate ℓ .
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Type curve.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent i type noise.
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization regime matrix.

Algorithm 5 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$.
- 2: **for** $i = 1, \dots, N$ **do**
- 3: Draw $u_i \sim \text{Unif}[0, 1]$, compute $\gamma(u_i) \in \mathbb{R}^d$, draw $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
- 4: $x_i^0 \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
- 5: Draw ideal type c as a noisy curve point.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

- 9: Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.
 - 10: Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire} ; add isolated broker node $b = N + 1$.
 - 11: Initialize all agent histories, partner means, MLPs, satisfaction scores, broker-tried flags, tenures, and active-match sets.
 - 12: Initialize broker history, broker MLP, reputation, and capture-confidence state.
 - 13: Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil 0.2N \rceil$; add broker edges (i, b) for all roster members.
 - 14: Seed $\mathcal{H}_b^0$ with up to 100 random unordered roster-member pairs; for each pair draw q_{ij} , append to $\mathcal{H}_b^0$, and add edge (i, j) .
 - 15: **for** $i = 1, \dots, N$ **do**
 - 16: Sample up to five non-broker neighbors j of i ; draw q_{ij} and append (x_j, q_{ij}) to $\mathcal{H}_i^0$; update $\bar{q}_{ij}^0, C_{ij}^0$.
 - 17: Set rep_b^0 to mean broker seed output; set $s_{i, \text{self}}^0$ to mean own seed output and $s_{i, \text{broker}}^0 \leftarrow \text{rep}_b^0$.
 - 18: Train each non-empty agent MLP on its seed history for E_{init} steps.
 - 19: Train the broker MLP on symmetry-augmented seed history for E_{init} steps.
 - 20: **return** S^0 .
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5 One Period

Algorithm 6 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .

Ensure: Updated state S^t and accepted records $\mathcal{R}^t$.

- 1: Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i , and set $D^t \leftarrow \emptyset$.
 - 2: Refresh roster: remove each roster member with probability p_{roster} , then replenish uniformly to size $\lceil 0.2N \rceil$.
 - 3: Synchronize broker edges to $\text{Roster}^t \cup D^t \cup \{i : M_i^t \text{ contains a broker-channel match}\}$.
 - 4: $\mathcal{D}^t \leftarrow \emptyset$.
 - 5: **for** $i = 1, \dots, N$ **do**
 - 6: $\tau_i^t \leftarrow \tau_i^{t-1} + 1$; draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
 - 7: **if** $d_i^t > 0$ **then**
 - 8: $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$.
 - 9: Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
 - 10: **if** $z_i^t = \text{broker}$ **then**
 - 11: Add i to D^t .
 - 12: Synchronize broker edges using the updated D^t .
 - 13: $\text{TRAINPREDICTORS}(S, t, \vartheta)$.
 - 14: $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$.
 - 15: $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$.
 - 16: **if** $D^t \neq \emptyset$ **then**
 - 17: $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$.
 - 18: **else**
 - 19: $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
 - 20: $\text{UPDATEBROKERCONFIDENCE}(S, \mathcal{R}^t)$ and compute pure diagnostics.
 - 21: **for** $i = 1, \dots, N$ **do**
 - 22: **if** agent i exits with probability η_{exit} **then**
 - 23: Replace slot i by a fresh entrant with new type, empty histories, fresh MLP, zero tenure, and $\lfloor k_G/2 \rfloor$ type-proximity edges.
 - 24: **if** $t \bmod \Delta_{\text{net}} = 0$ **then**
 - 25: Recompute broker betweenness, Burt constraint, and effective size on G^t .
 - 26: **return** $(S^t, \mathcal{R}^t)$.
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6 Offer Market

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Local variables for OfferMarket

Symbol	Dimension/type	Definition
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{Z}^t$	subset of $[N]$	Captured origins removed from standard offers.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.
$\mathcal{I}_b$	list	Broker pairs sorted by v_b^t .
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0, 1\}$	Return value of AddOffer.

Algorithm 7 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

- 1: **if** O_{ij}^t is defined **then**
 - 2: **return** 0.
 - 3: $O_{ij}^t \leftarrow (i, j, z, v)$.
 - 4: **if** O_{ji}^t is undefined **then**
 - 5: $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
 - 6: **return** 1.
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Algorithm 8 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .

Ensure: Accepted records $\mathcal{R}^t$.

- 1: For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
 - 2: $O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
 - 3: Draw one period-level stranger pool $U^t \subset [N]$ uniformly without replacement, $|U^t| = \min(n_{\text{strangers}}, N)$.
 - 4: $\mathcal{Z}^t \leftarrow \emptyset$.
 - 5: **if** principal mode is enabled **then**
 - 6: $\mathcal{Z}^t \leftarrow \text{CLIENTORIGINCAPTURE}(S, \mathcal{D}^t, \Omega, \mathcal{R}^t, \vartheta)$.
 - 7: **for** each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
 - 8: $\mathcal{U}_i^{\text{self}} \leftarrow \{j \in \mathcal{N}_G^t(i) : C_{ij}^t > 0\} \cup (U^t \setminus \mathcal{N}_G^t(i))$.
 - 9: Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t \cup \mathcal{Z}^t$.
 - 10: $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}: v_{i \rightarrow j}^{\text{self}, t} > r}(-v_{i \rightarrow j}^{\text{self}, t})$.
 - 11: **for** the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ agents j in $\mathcal{I}_i^{\text{self}}$ **do**
 - 12: $\text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t})$.
 - 13: $\mathcal{A}_b^t \leftarrow (\text{Roster}^t \cup D^t) \setminus \mathcal{Z}^t$.
 - 14: $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, i \notin \mathcal{Z}^t, \Omega_i > 0\}$.
 - 15: $\mathcal{P}_b^t \leftarrow \{\{i, j\} : i \neq j, (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$.
 - 16: $\mathcal{I}_b \leftarrow \text{argsort}_{\{i, j\} \in \mathcal{P}_b^t: v_b^t(i, j) > r}(-v_b^t(i, j))$.
 - 17: **for** each $\{i, j\} \in \mathcal{I}_b$ **do**
 - 18: **for** each ordered pair $(\alpha, \beta) \in \{(i, j), (j, i)\}$ **do**
 - 19: **if** $\alpha \in D_+^t$, $\beta \in \mathcal{A}_b^t$, and $\Omega_\alpha > 0$ **then**
 - 20: $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, \alpha, \beta, \text{broker}, v_b^t(i, j))$.
 - 21: $\Omega_\alpha \leftarrow \Omega_\alpha - \iota$.
 - 22: **for** each $\{i, j\} \in \mathcal{P}_O^t$ **do**
 - 23: $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
 - 24: **if** $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
 - 25: $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
 - 26: **else if** $o_{ij} \neq \emptyset$ and $w_{j \leftarrow i}^t > r$ **then**
 - 27: $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
 - 28: **else if** $o_{ji} \neq \emptyset$ and $w_{i \leftarrow j}^t > r$ **then**
 - 29: $\text{FINALIZESTANDARD}(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
 - 30: **return** $\mathcal{R}^t$.
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Algorithm 9 FinalizeStandard

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

- 1: **if** $j \in M_i^t$ **then**
 - 2: **return** .
 - 3: Draw $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
 - 4: Append (x_j^t, q_{ij}^t) to $\mathcal{H}_i$ and (x_i^t, q_{ij}^t) to $\mathcal{H}_j$.
 - 5: Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
 - 6: Add edge (i, j) to G^t ; add j to M_i^t and i to M_j^t .
 - 7: **if** $z = \text{broker}$ or $(o' \neq \emptyset \text{ and } o'.z = \text{broker})$ **then**
 - 8: Append (x_i^t, x_j^t, q_{ij}^t) once to $\mathcal{H}_b$.
 - 9: $z^* \leftarrow \text{broker}$.
 - 10: **else**
 - 11: $z^* \leftarrow \text{self}$.
 - 12: Append accepted record $(i, j, z^*, 0, q_{ij}^t, v, \text{NaN}, o, o')$ to $\mathcal{R}^t$.
-

7 Optional Client-Origin Resource Capture

Local variables for ClientOriginCapture

Symbol	Dimension/type	Definition
p_i	scalar	Acquisition price per captured position.
$\mathcal{J}_i$	subset of $[N]$	Planned recipient set.
Γ_i	scalar	Expected net capture gain.
$\mathcal{I}_c$	list	Feasible capture origins sorted by Γ_i .

Algorithm 10 ClientOriginCapture

Require: State S , demand records $\mathcal{D}^t$, residual quotas Ω , accepted records $\mathcal{R}^t$.

Ensure: Captured origins $\mathcal{Z}^t$; principal records appended to $\mathcal{R}^t$.

```
1: if  $n_{\kappa,b} < n_{\min}$  or  $\kappa_b^t/(\bar{q}_{\text{cal}} - r) > \kappa_{\max}$  then
2:   return  $\emptyset$ .
3:  $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$ ,  $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$ .
4: Score  $\mathcal{P}_b^t = \{\{i, j\} : i \in D_+^t, j \in \mathcal{A}_b^t, i \neq j\}$  by  $v_b^t(i, j)$ .
5: for  $i \in D_+^t$  do
6:    $p_i \leftarrow n_i^{-1} \sum_{\ell'=1}^{n_i} y_{i\ell'}$  if  $n_i > 0$ , else  $p_i \leftarrow \bar{q}_{\text{cal}}$ .
7:    $\mathcal{J}_i \leftarrow$  top  $\Omega_i$  distinct  $j \in \mathcal{A}_b^t \setminus \{i\}$  with  $v_b^t(i, j) > r$ .
8:    $\Gamma_i \leftarrow \sum_{j \in \mathcal{J}_i} v_b^t(i, j) - \Omega_i p_i - \Omega_i \phi$ .
9:  $\mathcal{I}_c \leftarrow \text{argsort}_{i: |\mathcal{J}_i| = \Omega_i, p_i \geq r, \Gamma_i > 0}(-\Gamma_i)$ ;  $\mathcal{Z}^t \leftarrow \emptyset$ .
10: for  $i \in \mathcal{I}_c$  do
11:   Recompute  $\mathcal{J}_i$  from the sorted pair list, excluding current matches and  $\mathcal{Z}^t$ .
12:   if  $|\mathcal{J}_i| < \Omega_i$  then
13:     continue.
14:   Add  $i$  to  $\mathcal{Z}^t$ ; record principal payment  $\Omega_i p_i$  to  $i$ ; set  $\Omega_i \leftarrow 0$ .
15:   for  $j \in \mathcal{J}_i$  do
16:     Mark  $(i, j)$  as current before receiver evaluation.
17:     if  $w_{j \leftarrow i}^t > r$  then
18:       Draw  $q_{ij}^t$ ; append  $(x_i^t, x_j^t, q_{ij}^t)$  to  $\mathcal{H}_b$ .
19:       Add principal active matches  $i \leftrightarrow j$ .
20:       Append  $(i, j, \text{broker}, 1, q_{ij}^t, v_b^t(i, j), p_i, \emptyset, \emptyset)$  to  $\mathcal{R}^t$ .
21:     else
22:       Record rejected principal exposure with realized value 0 and prediction  $v_b^t(i, j)$ .
23: return  $\mathcal{Z}^t$ .
```

Principal positions update broker history only when accepted. They do not update agent histories, agent-agent edges, or agent partner means.

8 Satisfaction, Confidence, and Metrics

Local variables for satisfaction and confidence

Symbol	Dimension/type	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}^{\text{std}}$	integer	Standard broker offer credits for demander i .
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
$P_i^{\text{capture},t}$	scalar	Principal payment to origin i , zero absent capture.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathcal{E}_{\text{err},b}^t$	set	Broker exposure-error observations.
e_b^t	scalar	Period mean absolute exposure error.

Algorithm 11 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

- 1: **for** each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
 - 2: $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, \zeta.\pi = 0\} \zeta.q + P_i^{\text{capture}, t}$.
 - 3: $n_{i,b}^{\text{std}} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}, \zeta.\pi = 0\}$.
 - 4: **if** $z_i = \text{self}$ **then**
 - 5: $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s$.
 - 6: **else**
 - 7: $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b}^{\text{std}})/d_i$; $\chi_i^t \leftarrow 1$.
 - 8: $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega\tilde{q}_i^t$.
-

Algorithm 12 UpdateBrokerConfidence

Require: State S , accepted records $\mathcal{R}^t$.

- 1: $\mathcal{E}_{\text{err},b}^t \leftarrow \emptyset$.
 - 2: **for** each accepted record $\zeta \in \mathcal{R}^t$ **do**
 - 3: **for** each standard offer credit $o \in \{\zeta.o_1, \zeta.o_2\}$ **do**
 - 4: **if** $o \neq \emptyset$ and $o.z = \text{broker}$ **then**
 - 5: Add $(o.v, \zeta.q)$ to $\mathcal{E}_{\text{err},b}^t$.
 - 6: **if** $\zeta.\pi = 1$ **then**
 - 7: Add $(\zeta.\hat{q}, \zeta.q)$ to $\mathcal{E}_{\text{err},b}^t$.
 - 8: **if** $|\mathcal{E}_{\text{err},b}^t| > 0$ **then**
 - 9: $e_b^t \leftarrow |\mathcal{E}_{\text{err},b}^t|^{-1} \sum_{(\hat{q}, q) \in \mathcal{E}_{\text{err},b}^t} |\hat{q} - q|$.
 - 10: $\kappa_b^t \leftarrow e_b^t$ if $n_{\kappa,b}^{t-1} = 0$, else $(1 - \omega)\kappa_b^{t-1} + \omega e_b^t$.
 - 11: $n_{\kappa,b}^t \leftarrow n_{\kappa,b}^{t-1} + |\mathcal{E}_{\text{err},b}^t|$.
-

CollectMetrics. COLLECTMETRICS($S^t, \mathcal{R}^t, \vartheta$) is a pure measurement function. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, degree summaries, cached broker network measures, and principal-mode diagnostics when enabled. Metrics do not feed back into S^{t+1} except through state variables updated above.